



A multi-functional binder for high loading sulfur cathode

Ying Chu^a, Ning Chen^a, Ximing Cui^a, Anmin Liu^c, Liang Zhen^b, Qinmin Pan^{a,*}

^aSchool of Chemistry and Chemical Engineering, Harbin Institute of Technology, Harbin 150001, Heilongjiang, China

^bSchool of Materials Science and Engineering, Harbin Institute of Technology, Harbin 150001, Heilongjiang, China

^cState Key Laboratory of Fine Chemicals, School of Petroleum and Chemical Engineering, Dalian University of Technology, Panjin 124221, Liaoning, China

ARTICLE INFO

Article history:

Received 30 August 2019

Revised 23 October 2019

Accepted 24 October 2019

Available online 7 November 2019

Keywords:

Li-S batteries

Sulfur cathode

Mussel-inspired binder

Bio-adhesion

High loading

High performance

ABSTRACT

Lithium sulfur (Li-S) batteries are the promising power sources, but their commercialization is significantly impeded by poor energy-storage functions at high sulfur loading. Here we report that such an issue can be effectively addressed by using a mussel-inspired binder comprised of chitosan grafted with catecholic moiety for sulfur cathodes. The resulting sulfur cathodes possess a high loading up to 12.2 mg cm⁻² but also exhibit one of the best electrochemical properties among their counterparts. The excellent performances are attributed to the strong adhesion of the binder to sulfur particles, conducting agent, current collector, and polysulfide. The versatile adhesion effectively increases the sulfur loading, depresses the shuttle effect, and alleviates mechanical pulverization during cycling processes. The present investigation offers a new insight into high performance sulfur cathodes through a bio-adhesion viewpoint.

© 2019 Science Press and Dalian Institute of Chemical Physics, Chinese Academy of Sciences. Published by Elsevier B.V. and Science Press. All rights reserved.

1. Introduction

Lithium sulfur (Li-S) batteries are the promising power sources for electrical vehicles (EVs) and hybrid electrical vehicles (HEVs) because of high theoretical energy-density, low cost, and environmental benignity of sulfur [1]. Unfortunately, the practical application of Li-S batteries is significantly hindered by fast capacity fading originated from the migration of polysulfide (which is also known as shuttle effect) and mechanical pulverization of sulfur particles. In the past decade, much effort had been devoted to addressing the shuttle effect by rationally confining polysulfide [2]. These attempts included embedding sulfur in host materials like carbonaceous matrix or polar polymer [3–5]. As for mechanical pulverization, nano-size sulfur is found to effectively relax the stress produced by volume expansion/contraction [6]. However, keeping excellent performance at a high loading remains a big challenge for sulfur cathodes [7]. It is imperative to increase the sulfur loading since high energy/power density is a crucial factor for practical applications.

As a key component for sulfur electrodes, binder plays a critical role in maintaining high loading and reliable performance [8–14]. In order to realize the above goals, a cathode should tightly bind sulfur particles while ensuring its integrity, electronic and ionic conductivity during cycling. However, such strong adhesion is

challenging for sulfur particles due to the absence of polar functionality, which greatly reduces the interaction between sulfur particles and the current collector or binder in liquid electrolyte. The situation becomes worse in the cycling process due to the dynamic volume change and even mechanical pulverization of sulfur particles. Besides strong adhesion, another challenge for a sulfur cathode is to effectively confine the migration of polysulfide. In order to realize both high-performance and high-loading, therefore, an ideal binder should have strong adhesion to sulfur particles, conducting agent, the current collector and polysulfide during the dynamic volume variation process. This means that a novel binder should exhibit strong adhesion to a wide range of solids and even molecular polysulfide, especially in liquid environments.

In this regard, marine mussels might offer an inspiration for designing a novel binder. Mussels exhibit outstanding adhesion properties even on wet surface by secreting adhesive proteins rich in catecholic moieties [15–25]. The mussel-inspired wet-adhesion is very valuable for Li-S batteries because tightly binding their electroactive components in liquid environments are crucial for reliable performance, especially at the high sulfur loading [26–28]. More importantly, the catecholic moiety might effectively hinder the migration of polysulfides by its polar hydroxyl and amino functionalities. All these suggest that the unique adhesion of catecholic moiety might offer a novel strategy to address the shuttle effect and high loading of sulfur cathodes.

Here, we design a mussel-inspired binder composed of chitosan grafted with catholic moieties for sulfur cathodes. The binder shows strong adhesion to sulfur particles, the current collector and

* Corresponding author.

E-mail address: panqm@hit.edu.cn (Q. Pan).

polysulfides, while its chitosan-based backbone buffers the volume change during the cycling process. Consequently, the resulting sulfur cathodes exhibit a high specific capacity and long cycling stability at a high sulfur loading. The present investigation provides a new insight into the high-performance sulfur cathodes through a bio-adhesion viewpoint.

2. Experimental

2.1. Materials and chemicals

Chitosan, dopamine hydrochloride (DA-HCl), benzyl bromide, vanillin, ethyl acetate, potassium carbonate, acetic acid, hydrobromic acid, aluminum chloride anhydrous, sodium borohydride (NaBH_4), pyridine, hydrochloric acid, sodium chloride, sodium sulfate, poly(vinyl alcohol) (PVA), 1,3-dioxolane (DOL), 1,2-dimethoxyethane (DME) and lithium bis(trifluoromethylsulfonyl) imide (LiTFSI) were purchased from Sinopharm Chemical Reagent Co., Ltd (China). Alumina foils with a thickness of 30 μm were provided by Luoyang Haotai Copper Co., Ltd. (China). TIMCAL SUPER P[®] Li, lithium foils and nickel foam were offered by Lizhiyuan Battery Sales Department (China) and China Energy Lithium Co., Ltd. (China), respectively.

2.2. Synthesis of the CNC binder

The CNC binder was synthesized through our previously reported processes. Briefly, chitosan substituted with nitrocatechol (chitosan-nitrocatechol, CNC) was synthesized through a reductive amination reported in the literatures (Fig. S1, Supporting Information) [29–31].

2.3. Preparation of sulfur cathodes and assembly of cells

Sulfur powder, super P and CNC binder with a mass ratio of 6:3:1 were mixed in deionized water to form a homogeneous slurry. The slurry was then coated on an alumina foil or nickel foam. After drying at 60 $^{\circ}\text{C}$, the resulting foil was punched into circles with a diameter of 1.5 cm. Coin cells were assembled in an argon-filled glove-box using the above circles as the working electrodes and lithium foils (with a thickness of 1 mm) as the counter electrodes. The electrolyte was 1.0 mol L^{-1} LiTFSI in DOL/DME with 2 wt% LiNO_3 as additive.

2.4. Electrochemical measurements

Cyclic voltammetry (CV) and electrochemical impedance spectroscopy (EIS) were measured on a CHI650d potentiostat. Galvanostatic charge/discharge (GCD) was conducted on a Neware battery tester (CT-4008-5 V50 mA-S1).

2.5. Shearing adhesion measurements

The CNC binder was first dissolved in water with a concentration of 50 mg mL^{-1} . The resulting solution was used to joint two aluminum alloy plates (1.0 mm in thickness). The contact area (A) between the two plates was 10.0 mm $\times$ 10.0 mm. After drying in vacuum, the plates were stretched until detachment and the force (F) were measured. The shearing adhesion strength (S, Pa) was determined by the formula $S = F/A$.

2.6. Measurements of the adhesion of CNC to sulfur powder and carbon black

At first, 0.3 g of PVA was dissolved into 10 mL of deionized water at 90 $^{\circ}\text{C}$ for 30 min. Then 0.1 g of CNC and 0.1 g of sulfur powder

(or super P) were added to the above solution under stirring for 2 h to obtain PVA-CNC-S or PVA-CNC-C solution, respectively. The solutions were coated on a polyfluorotetraethylene (PTFE) plate and drying at room temperature for 24 h. The resulting PVA-CNC-S or PVA-CNC-C film was peeled off and cut into strips of 40 mm $\times$ 5 mm $\times$ 0.1 mm. For comparison, the PVA-S or PVA-C films without CNC were prepared under the same conditions. Tensile experiments of the strips were performed on a universal electromechanical testing machine (Cree 8007B) at a rate of 50 mm min^{-1} . Each sample was tested for five times.

2.7. Computational details

The theoretical calculations of optimization and adsorption energy were carried out based on density functional theory (DFT) using the Perdew, Burke and Ernzerhof (PBE) functional of generalized gradient approximation (GGA) with the DNP basis set (basile_v4.4) in the Dmol3 package [32–34]. The optimization process was repeated until the total energy change of the system was less than 1.0×10^{-5} eV. The orbital cutoff of 4.40 Å and the thermal smearing of 0.005 Ha were employed to improve the computational performance. To evaluate the adsorption strength of $\text{Li}_2\text{S}/\text{LiS}^{\bullet}$ on the CNC binder, the adsorption energies ΔE_{ads} were calculated as

$$\Delta E_{\text{ads}} = - [E_{(\text{CNC}-\text{Li}_2\text{S}/\text{LiS}^{\bullet})} - E_{(\text{CNC})} - E_{(\text{Li}_2\text{S}/\text{LiS}^{\bullet})}]$$

where $E_{(\text{CNC})}$, $E_{(\text{Li}_2\text{S}/\text{LiS}^{\bullet})}$ and $E_{(\text{CNC}-\text{Li}_2\text{S}/\text{LiS}^{\bullet})}$ are the calculated electronic energies of the CNC binder, $\text{Li}_2\text{S}/\text{LiS}^{\bullet}$ and the anchoring systems, respectively [35]. The higher value of ΔE_{ads} indicates the stronger adsorption of $\text{Li}_2\text{S}/\text{LiS}^{\bullet}$ on the CNC binder.

2.8. Characterizations

Proton nuclear resonance ($^1\text{H-NMR}$) was conducted on a Bruker 400, and scanning electron microscopy (SEM) was performed on an FEI Quanta 200. Shearing adhesion measurements were conducted on a Cree 8007B. X-ray photoelectron (XPS) spectroscopy was measured on a PHI-5700ESCA. Fourier transform infrared spectroscopy (FT-IR) was performed on a Nicolet iS50. Ultraviolet visible (UV-Vis) spectroscopy was performed on a UV-6100 spectrometer (China). The theoretical calculations were carried out by the DMol3 module in the Materials Studio software (Accelrys Inc).

3. Results and discussion

At first, the catecholic-based (CNC) binder was synthesized through a reductive amination (Fig. S1, Supporting Information) [3,36]. The chemical structure of the binder was characterized by proton nuclear resonance ($^1\text{H-NMR}$), FT-IR and Raman spectroscopy, which shows the presence of catecholic moieties, polysaccharide backbone and nitro groups (Fig. 1(a) and Fig. S2 of Supporting Information). In the CNC binder, the catecholic moiety shows strong adhesion to solids in a wet environment [18,21,24]. Indeed, the wetness-resistant adhesion is very valuable for Li-S batteries because their components also operate in liquid electrolytes. On the other hand, the presence of nitro group can effectively impede the oxidation of catecholic functionality due to the electron-withdrawn nature [29,30]. Furthermore, recent investigations had confirmed the importance of the rigidity of polymer backbone in keeping cycling stability of electrode materials with mechanical pulverization like Si [37–39]. More importantly, the polar $-\text{OH}$, $-\text{NH}_2$ and $-\text{NO}_2$ groups generally show strong chemical or physical interactions with polysulfides [35,40–42]. Therefore, our CNC binder integrates the strong adhesion of catecholic moiety, the high rigidity of polysaccharide backbone, as well as the confining

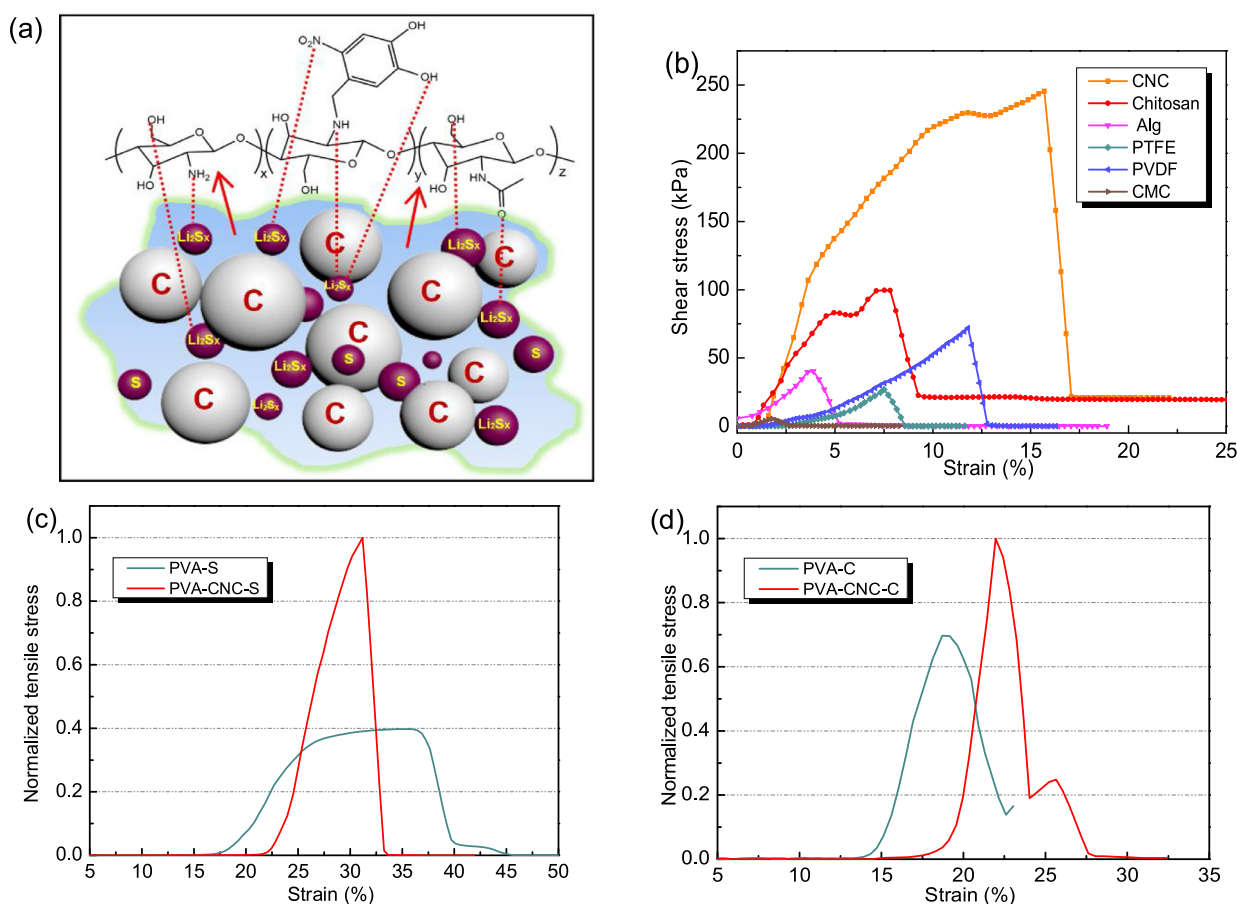


Fig. 1. (a) Illustration for the interactions between the polysulfides and CNC binder in the sulfur cathode. Inset is the chemical structure of the CNC binder. (b) Comparison on the shearing adhesion strength of the CNC and conventional binders. Effect of the CNC addition on the tensile strength of polyvinyl alcohol (PVA) films containing (c) sulfur powder and (d) carbon black (super P).

effect of polar functionalities, which is expected to be beneficial for the structural and electrochemical stability of a sulfur cathode during cycling (Fig. 1a).

The adhesive property of the CNC binder was then tested by shearing adhesion measurements. Here the binder, which was dissolved in water with a concentration of 50 mg mL⁻¹, was used to joint two Al alloy plates. After drying under vacuum, the plates were stretched until detachment. Fig. 1(b) is the typical shearing adhesion curves of the binder. It shows a shearing strength up to 245.4 kPa and a breaking strain of 15.7%. The strength value is much higher than that of pristine chitosan (i.e., 99.7 kPa). For comparison, we also measured the shearing adhesion strength of the conventional binders like polyvinylidene fluoride (PVDF, 71.9 kPa), sodium alginate (Alg, 40.3 kPa), polytetrafluoroethylene (PTFE, 26.9 kPa) and sodium carboxymethyl cellulose (CMC, 5.4 kPa). Under the same conditions, our CNC binder shows the highest shearing adhesion strength, indicating strong adhesion of the CNC binder to solid substrates. Moreover, a control experiment was conducted to confirm the strong adhesion of CNC to sulfur (S) and carbon (C) particles. Here sulfur particles and carbon black (super P) were embedded into polyvinyl alcohol (PVA) films in the presence or absence of CNC, and the resulting films were labeled as PVA-CNC-S, PVA-CNC-C, PVA-S and PVA-C, respectively. The films were then investigated by tensile experiments. Compared with its PVA-S (or PVA-C) counterpart, the PVA-CNC-S (or PVA-CNC-C) film shows a high tensile strength (Fig. 1c and d). These results demonstrated that the CNC binder had strong interactions with S and carbon black particles. The strong adhesion will effectively impede the detachment of sulfur particles from the current

collector in the cycling process, especially for the cathode with a high sulfur loading.

The resulting binder was employed to prepare sulfur cathodes. Here commercial micro-sized sulfur powder and carbon black (super P) were used as electrode materials and conducting agents, respectively. The as-prepared sulfur electrodes were assembled into coin cells and their electrochemical performances were evaluated by cyclic voltammetry (CV) and galvanostatic charge/discharge (GCD) measurements. The electrolyte used was 1.0 mol L⁻¹ LiTFSI in DOL/DME. Although the binder was soluble in water, it was stable in the electrolyte (Fig. S3, Supporting Information). The CV curves in Fig. 2(a) show anodic and cathodic peaks at 1.90, 2.18 and 2.51 V vs. Li/Li⁺, which can be ascribed to the insertion and de-insertion of lithium in sulfur particles, respectively. The CV curves also exhibit good reproducibility during multiple scanning processes, indicating high reversibility of the cathode. Except for the redox couples of sulfur, there are almost no other signals in the CV curves, implying the electrochemical inertness of the binder. Indeed, the binder was stable at the potential range of 1.7–2.8 V vs. Li/Li⁺ (Fig. S4, Supporting Information).

Fig. 2(b) is the typical voltage profiles of the cathode during galvanostatic charge and discharge at 0.2C. There are plateaus at the voltage ranges of 2.2–2.3 and 2.3–2.4 V vs. Li/Li⁺, corresponding to the anodic and cathodic peaks at the CV curves, respectively. The cathode delivers an initial charge capacity of 902.2 mAh g⁻¹ and discharge capacity of 855.5 mAh g⁻¹, exhibiting a coulombic efficiency of 94.8%. After charge-discharge for 100 cycles, the cathode still has a discharge capacity of 764.1 mAh g⁻¹, showing a retention rate of 89.3% (Fig. 2c). To confirm the role of catecholic

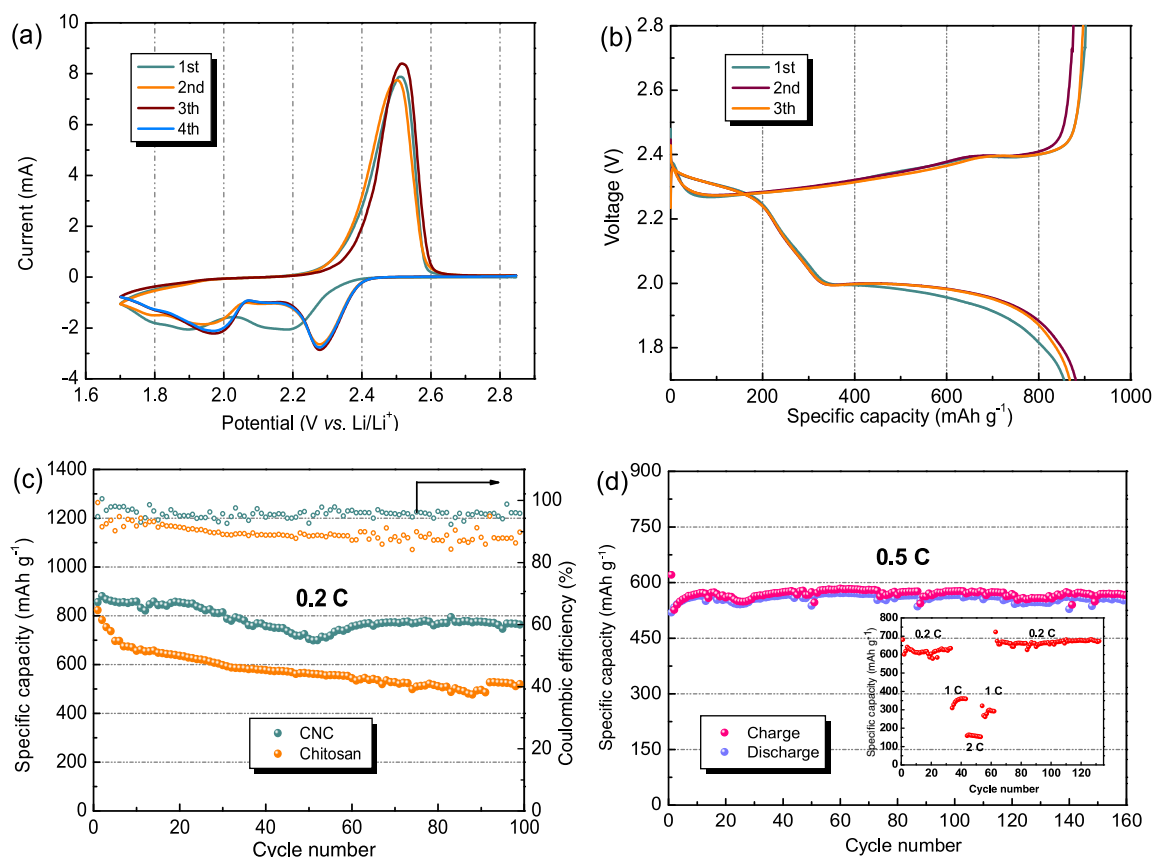


Fig. 2. Electrochemical performances of the sulfur cathodes used the CNC binder, (a) cyclic voltammograms at 0.5 mV s^{-1} , (b) GCD profiles and (c) cycling performance at 0.2 C , a control sulfur cathode used chitosan as binder was also measured under the same conditions. The sulfur loading was $\sim 2.0 \text{ mg cm}^{-2}$. (d) Cycling performance of the cathode with a sulfur loading of 4.0 mg cm^{-2} at 0.5 C . Inset is the rate capability of the cathode.

moiety in the electrochemical performance, we used pristine chitosan as the binder to prepare a control sulfur cathode. Under the same measurement conditions, the control sulfur cathode showed much lower specific capacity and cycling stability than its CNC-based counterpart (Fig. 2c). For instance, the control cathode only delivers a discharge capacity of 822.6 and 518.8 mAh g^{-1} at the first and the 100th cycles, respectively. The discrepancy confirms the importance of catecholic moiety on the electrochemical performances of sulfur electrodes. The difference is believed to arise from the relative low adhesion of chitosan compared with that of the CNC binder, although they contain the same polysaccharide backbone. Clearly, the presence of catecholic moiety greatly increases the adhesion of the binder to the current collector, sulfur particles and carbon black in the electrolyte. Consequently, the CNC binder is able to tightly bind sulfur particles onto the current collector, and thus effectively alleviates the structural collapse due to dynamic volume change of sulfur particles in the cycling process. As expected, the cathode shows good cycling stability at 0.5 C by delivering a specific capacity of 552.0 mAh g^{-1} at the 160th cycle (Fig. 2d). It also exhibits good rate capability at the current density of $0.5\text{--}2.0 \text{ C}$ (inset of Fig. 2d). Moreover, the CNC binder was beneficial for the cycling stability of a sulfur cathode using Al foil as the current collector (Fig. S5, Supporting Information).

Then we investigate the influence of sulfur loading on the electrochemical performances of the cathodes. Here sulfur electrodes with a loading ranging from 3.7 to 12.2 mg cm^{-2} were tested at a rate of 0.2 C (Fig. 3a). Under the same conditions, our electrodes could load sulfur particles up to 12.2 mg cm^{-2} , which is closed to those electrodes using guar gum and xanthan gum as binder [42]. Even at a high loading of 12.2 mg cm^{-2} , the electrode could keep

a stable capacity of 108.3 mAh g^{-1} after test for 98 cycles. The high loading and high cycling stability indicate that the binder can tightly bind sulfur particles and conducting agent to the current collector while maintaining good structural integrity, ionic conductivity and electrical contact. The strong adhesion was confirmed by the SEM observation performed on the sulfur cathode before and after 100 cycles (Fig. 3b and c). The images clearly show that the cycled cathode still keeps an integrated and compact morphology when compares with its original counterpart. The strong adhesion is attributed to the strong adhesion of catecholic moiety to sulfur particles, carbon black and the current collector. The adhesion mechanism arises from coordination, negative- π interaction, hydrogen bonding between the catecholic moiety and chitosan backbone [43]. Clearly, the high loading will greatly improve the energy-density (or power density) of sulfur cathodes, which might accelerate their practical application in real world.

To further investigate its confining effect on polysulfide, we mixed the CNC binder with Li_2S_6 in the DOL/DME solution. Here Li_2S_6 was used as a model compound to simulate polysulfide generated in the cathode. The strong interactions of the mussel-inspired binder with polysulfides were confirmed by the density-functional theory (DFT) calculations (Fig. S6 and Table S1, Supporting Information), X-ray photoelectron spectroscopy (XPS, Fig. 4), FI-IR spectroscopy (Fig. S7, Supporting Information), ultraviolet-visible (UV-vis) spectroscopy and visualized adsorption test (Fig. S8, Supporting Information). Fig. 4 is the XPS recorded on the resulting mixture after storage under ambient conditions. In the $\text{C } 1\text{s}$ spectrum, the peaks at 284.6 , 286.3 and 288.1 eV are assigned to the components of C-C , C-O/C-N=O and O-C-O/N-C=O in the CNC binder, respectively [44,45]. The peaks at 282.6 , 283.2 eV can be

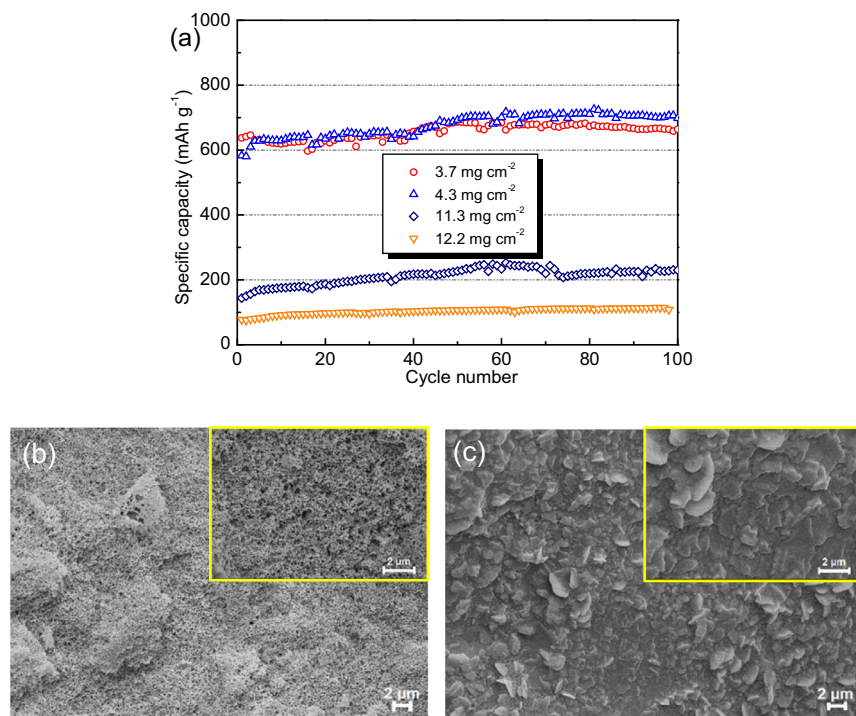


Fig. 3. (a) Cycling performances of the cathodes with different sulfur loadings at 0.2C. SEM images of the sulfur cathode (b) before cycling and (c) after test for 100 cycles at the rate of 0.2C.

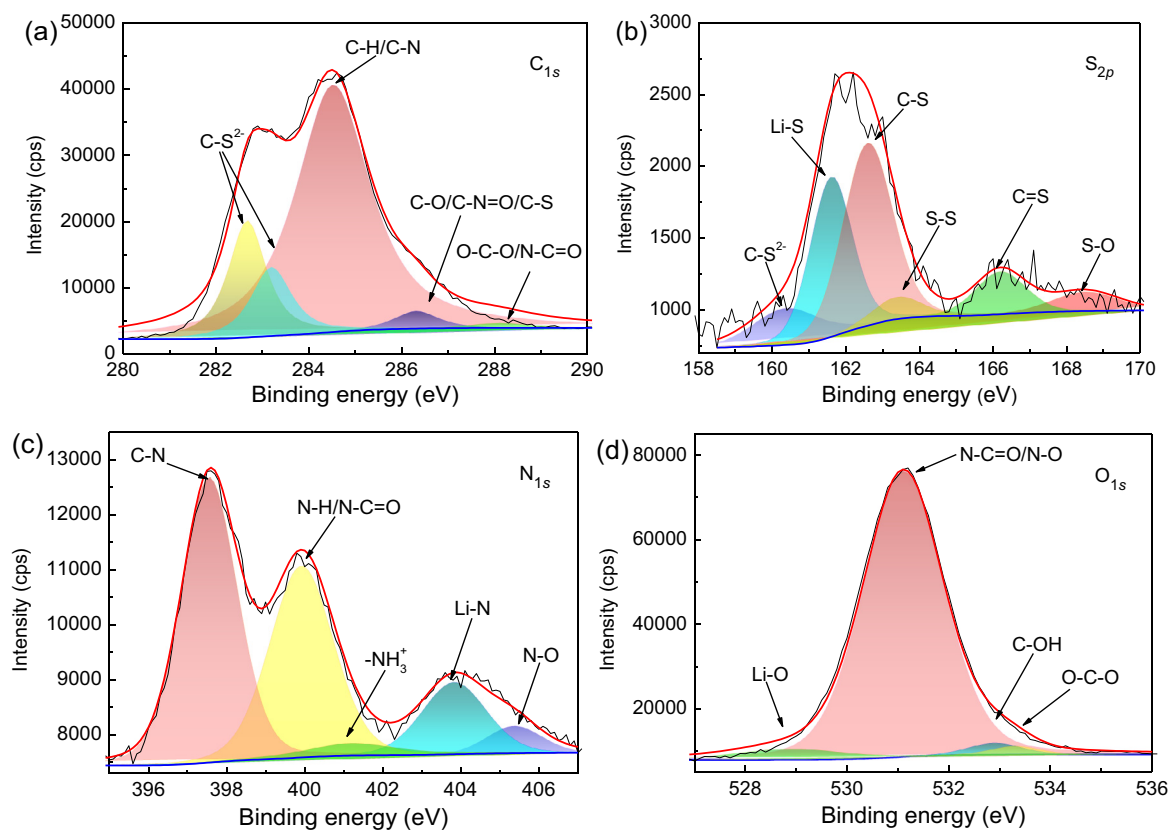


Fig. 4. X-ray photoelectron spectra show the interaction between the CNC binder and Li₂S₆ in the DOL/DME solution. (a) C 1s, (b) S 2p, (c) N 1s, (d) O 1s.

attributed to the $C-S^{2-}$ component originated from the interaction between the polysulfide and the CNC binder [46]. The S 2p spectrum in Fig. 4(b) also demonstrates the presence of $C-S^{2-}$ component at 160.4 eV. The existence of Li-S (161.6 eV) and C-S (162.6 eV) indicates the adsorption of the polysulfide by the CNC binder. The N 1s spectrum displays the peaks at 399.9 and 403.8 eV, corresponding to N-H/C-N=O and Li-N in the mixture, respectively. The peak at 401.2 eV is assigned to N-H bond in the ammonium form (NH_3^+). Both the $-NH_2$ and $-NH_3^+$ groups are likely to be present in the CNC binder (Fig. 4c) [12,47]. The O 1s peaks at 528.9, 531.1, 532.9 and 528.9 eV are assigned to Li-O, C-O, N-C=O, O-C-O, respectively (Fig. 4d). The above results suggest the strong interaction between the CNC binder and the polysulfide. The interaction lies in the fact that the lone pair electrons on the N atoms of the CNC binder coordinate with Li^+ of lithium polysulfides or Li_2S [48,49]. Therefore, the excellent performances of our sulfur cathodes are attributed to the strong adhesion of the CNC binder to sulfur microparticles, conducting agents, the current collectors and polysulfides. The versatile adhesion from macroscopic solids to molecular polysulfides ensures the sulfur cathodes keeping stable configuration integrity, structural stability and depressing the shuttle effect.

4. Conclusions

In summary, this study demonstrated that the energy-storage performances of sulfur cathodes with high sulfur loading were significantly enhanced by using a catechol-based binder. The resulting sulfur cathodes exhibited high specific capacity and cycling stability at high sulfur loading. The excellent performances were attributed to the strong adhesion of the binder to sulfur microparticles, conducting agent, the current collector and polysulfides in the electrolyte. The versatile adhesion is believed to arise from the presence of multiple functionalities that have chemical or physical interactions with these materials. The present investigation offers a new insight into high performances sulfur cathode via a bio-inspired strategy, which might be extended to other electrode materials with the similar issues.

Declaration of Competing Interest

The authors declare no conflict of interest.

Acknowledgments

The work was supported by the National Natural Science Foundation of China (51473041), the Foundation for Innovative Research Groups of the National Natural Science Foundation of China (51521003), China Postdoctoral Science Foundation (no. 2017M621262), Fundamental Research Funds for the Central Universities (No. HIT.NSRIF.201831) and Postdoctoral Foundation of Heilongjiang Province (LBH-Z17065).

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jechem.2019.10.020.

References

- [1] Y. Yin, S. Xin, Y. Guo, L. Wan, *Angew. Chem. Int. Ed.* 52 (50) (2013) 13186–13200.

- [2] S. Huang, R. Guan, S. Wang, M. Xiao, D. Han, L. Sun, Y. Meng, *Prog. Polym. Sci.* 89 (2019) 19–60.
- [3] H. Peng, Q. Zhang, *Angew. Chem. Int. Ed.* 54 (38) (2015) 11018–11020.
- [4] H. Wang, W. Zhang, J. Xu, Z. Guo, *Adv. Funct. Mater.* 28 (38) (2018) 1707520.
- [5] P.T. Dirlam, R.S. Glass, K. Char, J. Pyun, *J. Polym. Sci. Part A Polym. Chem.* 55 (10) (2017) 1635–1668.
- [6] X. Yang, X. Li, K. Adair, H. Zhang, X. Sun, *Electrochem. Energy Rev.* 1 (3) (2018) 239–293.
- [7] H. Peng, J. Huang, X. Cheng, Q. Zhang, *Adv. Energy Mater.* 7 (24) (2017) 1700260.
- [8] H. Chen, M. Ling, L. Hencz, H. Ling, G. Li, Z. Lin, G. Liu, S. Zhang, *Chem. Rev.* 118 (18) (2018) 8936–8982.
- [9] J. Liu, Q. Zhang, Y. Sun, *J. Power Sources* 396 (31) (2018) 19–32.
- [10] L. Li, T. Pascal, J. Connell, F. Fan, S. Meckler, L. Ma, Y. Chiang, D. Prendergast, B. Helms, *Nat. Commun.* 8 (2017) 2277.
- [11] F. Zeng, W. Wang, A. Wang, K. Yuan, Z. Jin, Y. Yang, *ACS Appl. Mater. Interfaces* 7 (47) (2015) 26257–26265.
- [12] G. Li, M. Ling, Y. Ye, Z. Li, J. Guo, Y. Yao, J. Zhu, Z. Lin, S. Zhang, *Adv. Energy Mater.* 5 (21) (2015) 1500878.
- [13] W. Chen, T. Lei, T. Qian, W. Lv, W. He, C. Wu, X. Liu, J. Liu, B. Chen, C. Yan, J. Xiong, *Adv. Energy Mater.* 8 (12) (2018) 1702889.
- [14] W. Chen, T. Qian, J. Xiong, N. Xu, X. Liu, J. Liu, J. Zhou, X. Shen, T. Yang, Y. Chen, C. Yan, *Adv. Mater.* 29 (12) (2017) 1605160.
- [15] Q. Qi, X. Lv, W. Lv, Q. Yang, *J. Energy Chem.* 39 (2019) 88–100.
- [16] H. Yuan, J. Huang, H. Peng, M. Titirici, R. Xiang, R. Chen, Q. Liu, Q. Zhang, *Adv. Energy Mater.* 8 (31) (2018) 1802107.
- [17] D.S. Wu, F. Shi, G. Zhou, C. Zu, C. Liu, K. Liu, Y. Liu, J. Wang, Y. Peng, Y. Cui, *Energy Storage Mater.* 13 (2018) 241–246.
- [18] H. Lee, S.M. Dellatore, W.M. Miller, P.B. Messersmith, *Science* 318 (5849) (2007) 426–430.
- [19] H. Lee, N.F. Scherer, P.B. Messersmith, *P. Natl. Acad. Sci.* 103 (35) (2006) 12999–13003.
- [20] C.V. Benedict, J.H. Waite, *J. Morphol.* 189 (2) (1986) 171–181.
- [21] H. Silverman, F. Roberto, *U. Marine Biotech.* 9 (6) (2007) 661–681.
- [22] H. Lee, B. Lee, P. Messersmith, *Nature* 448 (2007) 338–342.
- [23] J. Sedo, J. Saiz-Poseu, F. Busque, D. Ruiz-Molina, *Adv. Mater.* 25 (5) (2013) 653–701.
- [24] Q. Ye, F. Zhou, W. Liu, *Chem. Soc. Rev.* 40 (2011) 4244–4258.
- [25] M. Lyng, R. Westen, A. Postma, B. Städler, *Nanoscale* 3 (12) (2011) 4916–4928.
- [26] M. Ryou, J. Kim, I. Lee, S. Kim, Y. Jeong, S. Hong, J. Ryu, T. Kim, J. Park, H. Lee, J. Choi, *Adv. Mater.* 25 (11) (2013) 1571–1576.
- [27] M. Ryou, D. Lee, J. Lee, Y. Lee, J. Park, J. Choi, *Adv. Energy Mater.* 2 (6) (2012) 645–650.
- [28] M. Ryou, Y. Lee, J. Park, J. Choi, *Adv. Mater.* 23 (27) (2011) 3066–3070.
- [29] S.C. Tsai, J.P. Klinman, *Bioorg. Chem.* 31 (2) (2003) 172–190.
- [30] D.A. Learmonth, P.C. Alves, *Synth. Commun.* 32 (4) (2002) 641–649.
- [31] N. Chen, L. Qin, Q. Pan, *J. Mater. Chem. A* 6 (15) (2018) 6667–6674.
- [32] J. Bakera, A. Kessi, B. Delley, *J. Chem. Phys.* 105 (192) (1996) 192–212.
- [33] J.P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* 77 (18–28) (1996) 3865–3868.
- [34] J. Andzelm, R.D. King-Smith, G. Fitzgerald, *Chem. Phys. Lett.* 335 (3–4) (2001) 321–326.
- [35] Z. Ji, B. Han, Q. Li, C. Zhou, Q. Gao, K. Xia, J. Wu, *J. Phys. Chem. C* 119 (35) (2015) 20495–20502.
- [36] N. Chen, Q. Pan, *ACS Sustain. Chem. Eng.* 5 (9) (2017) 7905–7911.
- [37] I. Kovalenko, B. Zdyrko, A. Magasinski, B. Hertzberg, Z. Milicev, R. Burtovyy, I. Luzinov, G. Yushin, *Science* 334 (0526) (2011) 75–79.
- [38] N.S. Choi, K.H. Yew, W.U. Choi, S.S. Kim, *J. Power Sources* 177 (2) (2008) 590–594.
- [39] A. Magasinski, B. Zdyrko, I. Kovalenko, B. Hertzberg, R. Burtovyy, C. Huebner, T. Fuller, I. Luzinov, G. Yushin, *ACS Appl. Mater. Interfaces* 2 (11) (2010) 3004–3010.
- [40] W.S. She, Q. Zhang, W. Li, G. Zheng, H. Yao, Y. Cui, *Chem. Sci.* 4 (9) (2013) 3673–3677.
- [41] J. Song, T. Xu, M.L. Gordin, P. Zhu, D. Lv, Y. Jiang, Y. Chen, Y. Duan, D. Wang, *Adv. Funct. Mater.* 24 (9) (2014) 1243–1250.
- [42] J. Liu, D.G.D. Galpaya, L. Yan, M. Sun, Z. Lin, C. Yan, C. Liang, S. Zhang, *Energy Environ. Sci.* 10 (3) (2017) 750.
- [43] Q. Zhu, Q. Pan, *ACS Nano* 8 (2) (2014) 1402–1409.
- [44] G.D.B. Beamson, *High Resolution XPS of Organic Polymers*, Wiley, Chichester, 1992.
- [45] I.F. Amaral, P.L. Granja, M.A. Barbosa, *J. Biomater. Sci. Polym. Ed.* 16 (12) (2005) 1575–1593.
- [46] Y. Pang, J. Wei, Y. Wang, Y. Xia, *Adv. Energy Mater.* 8 (10) (2018) 1702288.
- [47] W. Chen, T. Lei, T. Qian, W. Lv, W. He, C. Wu, X. Liu, J. Liu, B. Chen, C. Yan, J. Xiong, *Adv. Energy Mater.* 8 (12) (2018) 1702889.
- [48] L. Yin, J. Liang, G. Zhou, F. Li, R. Saito, H. Cheng, *Nano Energy* 25 (2016) 203–210.
- [49] W. Zhou, X. Xiao, M. Cai, L. Yang, *Nano Lett.* 14 (9) (2014) 5250–5256.